

Assignment Day – 4

1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.



Steps I followed to reseat the RAM:

1. Shut down the laptop and unplugged the charger.
2. Removed the back cover carefully.
3. Located the RAM slot on the motherboard.
4. Pushed the side clips outward to release the RAM module.
5. Removed the RAM module by holding it from the edges.
6. Checked the RAM slot and module for dust or visible damage.
7. Aligned the notch on the RAM with the slot and reinserted it at an angle.
8. Pressed it down until the side clips locked it in place.

9. Reattached the back cover and powered on the laptop.

10. The laptop booted normally and the RAM was detected.

Issues observed:

No major issues were observed. The RAM clips were slightly tight, so gentle pressure was required.

2. **Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.**

Hint: Think about display issues, driver checks, and physical connections.



GPU Troubleshooting Checklist

1. **Identify the symptom – Check for no display, screen artifacts, crashes, or poor graphics performance.**
2. **Check display connections – Make sure the monitor cable is connected securely to the GPU output port and the monitor is set to the correct input.**
3. **Check the GPU physically – Power off the PC and ensure the GPU is properly seated in the PCIe slot.**
4. **Check GPU power cables – Make sure all required PCIe power connectors from the PSU are firmly connected.**
5. **Check drivers – Open Device Manager and check for GPU errors. Update or reinstall the graphics driver if necessary.**
6. **Test the GPU – Restart the PC and run a graphics-intensive application or benchmark to check for crashes, artifacts, or overheating.**
7. **Confirm the fix – If the display works normally and the system runs graphics applications without errors, the GPU issue is resolved.**

3. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.



Steps and observations:

1. Shut down the test PC and disconnect the power.
2. Unplugged the SATA data/power cable from the SSD/HDD.
3. Powered on the PC and entered the BIOS/UEFI.

- 4. Before reconnecting: The storage drive was not listed under Storage/SATA Information. The system may also show “No Boot Device Found” or “Boot Device Not Found.”**
- 5. Shut down the PC and reconnected the SATA cables securely.**
- 6. Powered on the PC and checked the BIOS again.**
- 7. After reconnecting: The SSD/HDD appeared in the BIOS storage list, and the PC was able to detect the boot drive and start Windows normally.**

4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.



Component to check first: RAM

I would check RAM first because faulty, loose, or incompatible RAM can cause system instability, crashes, and random restarts. It is also relatively easy to inspect and test.

Diagnostic steps:

1. Power off the PC and unplug it.
2. Open the case and check that the RAM modules are properly seated in their slots.
3. Remove and reseat the RAM firmly.
4. If there are multiple RAM sticks, test one stick at a time to identify a faulty module.
5. Run Windows Memory Diagnostic to check for memory errors.
6. If errors are found, test with a known-good compatible RAM module.
7. Restart and use the PC normally to confirm whether the random restarts have stopped.